

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of B. Tech. (CE/IT/EC/CSE) Theory Examination****November-December 2018****ME144 Elements of Engineering****Date: 26/11/2018, Monday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I must be attempted in drawing sheets and Section II must be attempted in answer sheets.
3. Make suitable assumptions and draw neat figures wherever is required.
4. Figures drawn in the question paper are not to the scale.
5. Show all the necessary dimensions in the solution.
6. Figures to right indicate the full marks.
7. Stencil work is not compulsory.

SECTION – I**Q - 1 Do as directed.**

1. What is the full form of A.S. in section of solid? [01]
a) Apparent b) Auxiliary Section c) Auxiliary Shape d) None of these
Section
2. Which of the following is not a type of scale? [01]
a) Plain Scale b) Diagonal Scale c) Objective Scale d) Vernier Scale
3. Which of the following is used for the identification of sheet? [01]
a) Title block b) T-square c) Drafter d) None of these
4. In first angle projection system, the object is kept in first quadrant. True or False? [01]
5. Isometric drawing is used to draw a 3D object from 2D views by _____. [01]
a) Computer b) Information c) Electronics & d) All of these
Engineers Technology Communication
Engineers Engineers
6. Engineering Graphics is the universal language and tool of communication for engineers. [01]
True or False?
7. Fusion 360 is a CAD software. True or False? [01]

Q – 2.a Draw an ellipse having major axis as 120 mm and minor axis as 80 mm using any method of the ellipse. Draw normal and tangent at any point on the curve. [06]

Q – 2.b A square prism side of base 30 mm and height 60 mm is resting on HP on one of the edges of the base. Rectangular face containing that edge on which rests on HP makes an angle 45° with HP. The side on which it rests on HP makes 30° with VP. Draw the projections of the prism. [06]

OR

Q – 2.b Classify the solids with necessary sketches. [06]

Q – 2.c Discuss the advantages of CAD software. [04]

Q – 3.a **Figure 1** shows a pictorial view of an object. Draw the following views: (1) Front View, [06]
(2) Right Hand Side View. Dimensions shown in the figure are in mm.

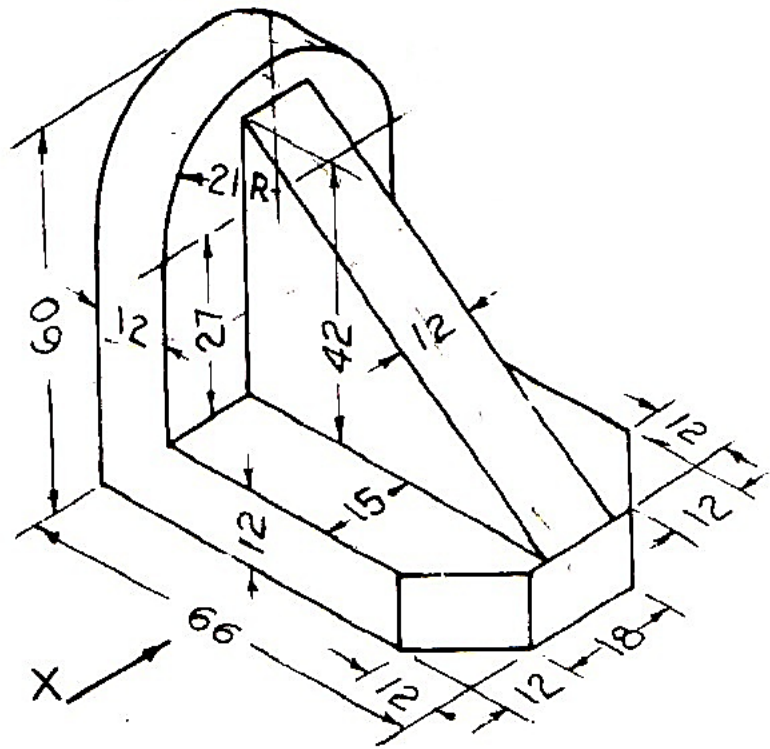


Figure 1

Q – 3.b A cylinder has a diameter of base 50 mm and height of axis 100 mm long, is resting on its base on HP. Draw the isometric drawing of the cylinder. [06]

OR

Q – 3.b Figure 2 shows the orthographic projections of an object in a first angle projection system. Draw its isometric view. Dimensions shown in the figure are in mm. [06]

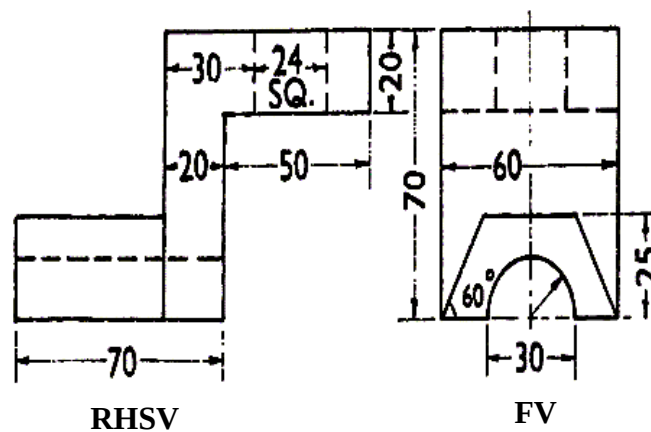


Figure 2

SECTION – II

Q - 4 Choose one correct alternative of the following multiple-choice questions.

1. The main scope of civil engineer is _____.

[01]

- | | | | |
|---------------|----------------|-------------------|-----------------|
| a) Planning, | b) Supervising | c) Maintenance of | d) All of these |
| designing and | construction | structures | |

estimating

2. _____ includes soil investigation, soil testing and foundation design. [01]
 a) Construction Planning & Management b) Transportation Engineering c) Geotechnical Engineering d) Structural Engineering
3. Which of the following is not a type of shallow foundation? [01]
 a) Spread footing b) Raft foundation c) Pile foundation d) Strap footing
4. _____ includes the curvature of the earth. [01]
 a) Geodetic surveying b) Plane surveying c) Solid surveying d) None of these
5. Which of the following is the type of building based upon occupancy? [01]
 a) Residential building b) Educational building c) Mercantile building d) All of these
6. Door should be located at _____ as per principal of planning-Privacy. [01]
 a) Any location b) Corner c) Center d) None of these
7. Which of the following is not considered as a design load? [01]
 a) Traffic load b) Dead Load c) Wind load d) Live load

Q – 5.a Draw a scale of 1:60 to show meters and decimeters and long enough to measure up to 6 meters. Show 3.7 meter on a plain scale. [06]

Q – 5.b Compare load bearing structure with framed structure. [04]

OR

Q – 5.b Enlist the principle of building planning and explain any one in detail. [04]

Q – 6.a Answer the following questions.

1. State the zeroth law and first law of thermodynamics. [03]
2. Derive combine gas law equation. [05]
3. State advantages of split air conditioning system over window air conditioning system. [03]

Q – 6.b Explain working of four-stroke petrol engine with neat sketches. [07]

OR

Q – 6.b Describe working and construction of simple vertical boiler with a neat sketch. [07]
